EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 3, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-03 **Date:** August 27, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Let d be a positive integer. If

$$a_1 \equiv r_1 \pmod{d} \quad \text{and} \quad a_2 \equiv r_2 \pmod{d},$$

then prove that the following theorem holds:

$$a_1 a_2 \equiv r_1 r_2 \pmod{d}.$$

[CO3, C2] [Mark: 5]

2. Determine the Big-O for each of the following functions. Simplify your answer to the tightest (most precise) Big-O expression.

(a) $(n^2 + 8)(n + 1)$

(b) $(n \log n + n^2)(n^3 + 2)$

(c) $(n! + 2^n)(n^3 + \log(n^2))$

[CO3, C2, Mark: 5]

EAST WEST UNIVERSITY

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B.Sc. in Computer Science and Engineering Program

Class Test 3, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-09 **Date:** August 28, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Generate 5 pseudorandom numbers using the Linear Congruential Generator method with the following parameters:

$$\text{Modulus } (m) = 12, \quad \text{Multiplier } (a) = 3, \quad \text{Increment } (c) = 5, \quad \text{Seed } (X_0) = 8$$

[CO3, C2] [Mark: 5]

2. Determine the Big-O for each of the following functions. Simplify your answer to the tightest (most precise) Big-O expression.

(a) $f(n) = 3n \log(n!) + (n^2 + 3) \log n$

(b) $f(n) = (n + 1) \log(n^2 + 1) + 3n^2$

[CO3, C2, Mark: 5 (3+2)]

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Class Test 3, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-10 **Date:** August 28, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Find the prime factorization of 2310399. Show the necessary steps.
[CO3, C2] [Mark: 5]

2. Find the average case time complexity of linear search. The algorithm is given below.

Algorithm 1 Linear Search

```
1: loc = -1
2: for i = 0 to n − 1 do
3:     if A[i] == key then
4:         loc = i
5:     end if
6: end for
```

[CO3, C2, Mark: 5]